QPCode

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP BCA (HONOURS) EXAMINATION APRIL 2025
SECOND SEMESTER
MG2CCRBCA102 OPERATING SYSTEMS
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create(C), Skill(S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer all questions
Each question carries **2** marks

1. Define the user view of an operating system. [K] [CO1]
2. What is the purpose of a Process Control Block (PCB)? [U] [CO2]
3. What are the benefits of the Co-operating Process? [U] [CO3]
4. What is internal fragmentation? [U] [CO4]
5. What is a page fault in demand paging [U] [CO4]

[2X5 =10]

Part B
Short Answer Questions
Answer any **5** questions
Each question carries **6** marks

6. Explain the working of a simple operating system structure. [U] [CO1]
7. Explain the different states of a process with a diagram. [U] [CO2]
8. Explain the advantages and disadvantages of multithreading. [U] [CO2]
9. What are semaphores? Explain two primitive semaphore operations. What are its advantages? [An] [CO3]
10. Define system call. Explain the importance of system calls in an operating system. [K] [CO1]
11. Describe how semaphores are used to solve the critical-section problem. [U] [CO3]
12. What is paging, and why is it used in memory management? [U] [CO4]

[6X5 =30]

Part C
Essay Questions
Answer any **2** questions
Each question carries **15** marks

13. Explain various process scheduling algorithms with suitable examples [U] [CO2]
14. Describe the different methods of interprocess communication. [U][CO3]
15. Explain the characteristics and workings of FIFO, LRU, and Optimal page replacement algorithms with examples. [A][CO4]

[15X2 =30]